Individual Assignment Cover Sheet

MBA 2024/2025

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1. Solution Outline

Our team solution specifies Green Data's strategy for achieving sustainable data centre operations, focusing on AI, alternative energy, and efficient infrastructure. Our key points were:

- *AI for Sustainability* - We exposed that AI-driven optimisations aim to reduce emissions and energy consumption in data centres. AI-powered fleet management and predictive maintenance for driverless vehicles improve energy efficiency, while AI-driven workload distribution in data centres minimises power waste.
- *Alternative Energy Sources* - The plan addressed transitioning data centres to renewable energy. Solar power is prioritised due to cost-effectiveness, with co-investment in solar farms in southern England. Additional renewable options, including tidal and wind power, are considered for scalability.
- *Strategic Data Centre Planning* - Our proposal included deploying underwater data centres to leverage natural cooling, reducing energy consumption. AI-enhanced workload management will dynamically optimise computing resources, while smart cooling systems and heat reuse will cut water and power use.

Green Data partners with SAC and ARM to accelerate renewable energy adoption and energy-efficient AI hardware integration. The overarching goal is to achieve net-zero emissions by 2030 through our proposed sustainable initiatives, and a long-term commercial plan up to 2034.

2. Discussion

Context

Our project was based framework for our analysis for the Green Data challenge was introduced during our second lecture (Barret, 2025). We considered the capabilities of ARM across relevant industries including Driverless Cars and low power chips. Consequently, it was relevant for us to consider using their technologies for the datacentre of our hypothetical company. Renewable solar energies provided by SAC (Alves, 2025) was the second commercial consideration.

Green Data had a major objective: reducing the environmental impact of their data centres specially those executing the AI algorithmic planning for driverless cars. Such ambitious objective for agile data driven organisation ready to innovate at high scale and fast pace is defined as a Massive-Transformative-Purpose (MTP) for Green Data (Ismail et al, 2014). The key reason to use AI in the driverless vehicles was focused to improved safety, reduced traffic congestion, and enhanced fuel efficiency (Tegmark, 2017).

Our business plan considered Green Data readiness for the upcoming innovations in transportation and prepare for global expansion while considering the technological trends and market forces (Osterwalder and Pigneur, 2010). For instance, the major economies besides UK, such as US and China are investing significantly into new technologies applied on roads, energy efficiency and AI implementation in driverless vehicles. Some of the benefits will be reducing energy consumption and road fatalities (Lee, 2018).

Environmental Impact challenges

According to relevant tech trends, Cloud Computing growth among top players is decreasing but investment on GenAI is increasing significantly together with overall R&D for those players (CB Insights, 2023). This helps to understand that Green Data investment on data servers would be focused mainly on the GenAI and AI powering the driverless cars rather than only using cloud services.

Based on the focus on GenAI for our company, there are some key facts to consider about growth of energy demand for the AI data centres and hence environmental impact. On one hand it is claimed that some AI tasks do not consume too much energy. For instance, AI writing a page of text and generating images emit 130x and 310x less CO2 respectively than a human created material. DeepMind reduces Google's energy consumption in data centres by 30% (Tomlinson et al, 2023).

On the other hand, there 8000 data centres in the World and energy consumption is expected to double from 2022 to 2026. Training an GenAI model generates the same CO2 than 5 standard cars during their whole lifespan (Marr, 2024). In terms of water, Google had a major controversy in Chile for a data centre pulling 170 litres/second in a populated area prone to draughts (DW News, 2024).

Resourcefully, ARM has addressed those challenges through high energy-efficient processors. This was inspired based on data centre cases in Ireland, US (Ashton, 2023), while countries like Singapore restricting construction of those centres.

Cooling and energy opportunities

We decided to work with SAC, based on their successful implementation on building a 5MW solar farm (Alves, 2025). A challenge would be UK's reduced sunlight availability, so we adjusted our calculations for our facility in Southern UK with relative lower solar exposure but bigger facility to reduce 70k-tonnes CO2 emissions for 2024-2034. The case of the Horsey Levels solar farm is a successful case of a large solar farm in the South of England just 4 miles away from the sea (National Grid. (2024).

Another opportunity identified was the placing of our data centre underwater and in the proximity solar farm situated by the coast. A successful case by Microsoft took place in Scotland, a data centre 117 feet deep to the seafloor (Microsoft, 2020). This would mean for our data centre: efficient cooling and less losses on energy transmission while close to the farm.

Data Centres and AI

The usage of natural water streams and AI algorithms to reduce environmental impact is another opportunity. For instance, Meta aiming to reduce their net emissions of 7.4M tonnes of CO2, uses river resources with fish passages to cool down their data centres (Climate Insider, 2024). In terms of AI, Meta is using Reinforcement Learning methods to optimise the allocation of energy and cooling resources (InfoQ, 2024). This goes in accordance with how AI systems can predict and adjust to changing demands, leading to reduced energy consumption and costs (Tegmark, 2017).

New electronic design is another opportunity we can leverage. AI can work locally in a device reducing energy consumption (Clark, 2024). ARM new chip designs applied on low energy consumption (Fotheringham and Teich, 2021) can be used on both the vehicles and the data centres.

Conclusion

Given the opportunities and challenges presented above, we proposed a plan for Green Data to minimise data centre environmental impact, especially for AI in driverless cars. Using ARM's low-power chips and SAC's solar energy, we explored sustainable cooling, AI-driven optimisation, and innovative electronic designs to reduce CO2 emissions and energy consumption efficiently.

References

Ashton, J. (2023) *The Everything Blueprint: The Microchip Design that Changed the World*. London: Hodder & Stoughton. pp. 283-284.

Alves, T. (2025) *The Digital Challenge – Session 4: Develop a Sustainable IT Proposition for Green Data*. Solar Americas.

Barret. M. (2025). *Creating Value through Digital Platforms for Ecosystem Advantage* [Lecture]. *MBA116 Digital Business*. Lecture Slides Session 2. University of Cambridge, Judge Business School. 23 January 2025.

CB Insights (2023) *Big Tech: The Future in 40 Charts.* Available at: https://www.cbinsights.com/research/report/big-tech-future-charts/ (Accessed: 16 February 2025).

Clark, D. (2024) *Artificial Intelligence for Learning: Using AI and Generative AI to Support Learner Development*. 2nd edn. London: Kogan Page. pp. 251-252.

Climate Insider (2024) *Key takeaways from Meta's 2024 sustainability report. Available* at: https://climateinsider.com/2024/09/03/key-takeaways-from-metas-2024-sustainability-report/ (Accessed: 16 February 2025).

DW News (2024) *How AI causes serious environmental problems (but might also provide solutions).* 15 May. Available at: https://www.youtube.com/watch?v=9aBNIekp4qY (Accessed: 16 February 2025).

Fotheringham, V. & Teich, P. (2021) *Energy-efficient data centers: How ARM processors are shaping the future of cloud computing*. IEEE Transactions on Sustainable Computing, 6(3), pp. 215-229.

InfoQ (2024). *Meta Optimizes Data Center Sustainability with Reinforcement Learning*. Available at: https://www.infoq.com/news/2024/10/data-center-sustainability-ai/ (Accessed: 16 February 2025).

Ismail, S., Malone, M. and Van-Geest, Y. (2014), *Exponential organizations: why new organizations are ten times better, faster, and cheaper than yours (and what to do about it)*, Diversion Books, New York. pp. 54-57.

Lee, K.-F. (2018) *AI Superpowers: China, Silicon Valley, and the New World Order*. Boston: Houghton Mifflin Harcourt. pp. 132-136.

Marr, B. (2024) *Generative AI in Practice: 100+ Amazing Ways Generative Artificial Intelligence is Changing Business and Society*. Hoboken, NJ: John Wiley & Sons. pp. 64-68.

Microsoft (2020). *Project Natick: Underwater Datacenter.* Available at: https://news.microsoft.com/source/features/sustainability/project-natick-underwater-datacenter/ (Accessed: 16 February 2025).

National Grid. (2024). *Horsey Levels solar farm takes home Solar & Storage gold after accelerated connection*. National Grid. Retrieved from https://www.nationalgrid.co.uk/news-and-events/latest-news/horsey-levels-solar-farm-takes-home-solar-storage-gold-after-accelerated-connection (Accessed 9 Feb. 2025).

Osterwalder, A. and Pigneur, Y. (2010) *Business Model Generation: A Handbook for Visionaries, Game Changers, and Challengers*. Hoboken, NJ: John Wiley & Sons. pp. 200-211.

Tegmark, M. (2017). *Life 3.0: Being Human in the Age of Artificial Intelligence*. New York: Alfred A. Knopf. pp. 130-132.

Tomlinson, B., Black, R.W., Patterson, D.J. and Torrance, A.W. (2023) *The carbon emissions of writing and illustrating are lower for AI than for humans*. Available at: https://arxiv.org/abs/2303.06219 (Accessed: 17 February 2025).